Machine Learning - Time series

Lecture #7 – 2023-12-19

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ALBERTSCHOOL

Machine Learning - Time Series

Time Series
Times series data consists in **features + past target values + time itself**.

Importance of time series data in various domains (finance, meteorology, economics, etc.).
Example:
- Weather data, rainfall measurements, temperature
- Heart rate monitoring (EKG), Brain monitoring (EEG)
- Quarterly sales, stock price, industry forecast, interest rate

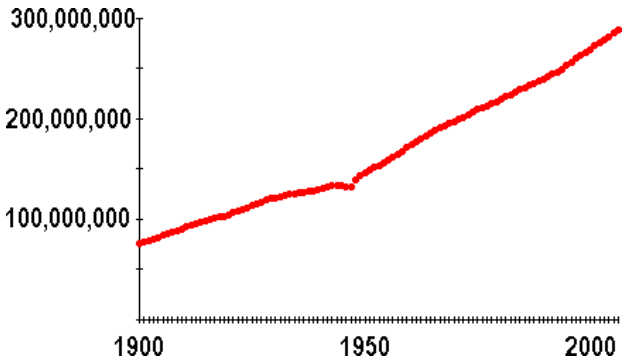
Unique characteristics of time series data (trend, seasonality, cyclicality)

WHAT ARE TIME SERIES ?

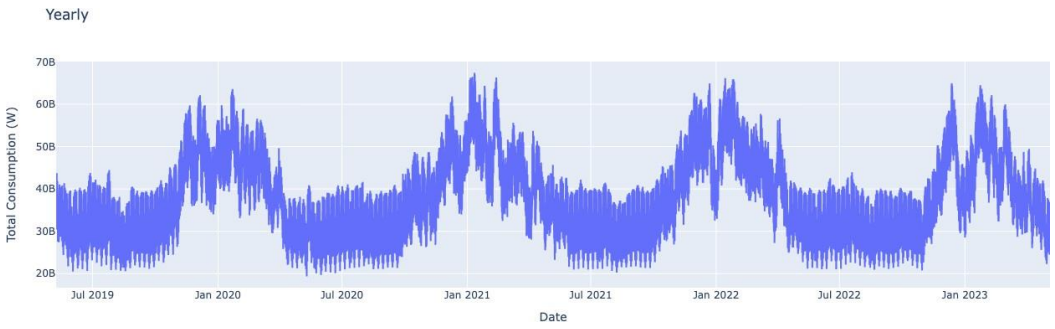
A time series is a **sequence** of **numerical data** points in **successive order**.
A stochastic process $\{\ldots, X_{t-1}, X_t, X_{t+1}, \ldots\}$ consisting of random variables indexed by time index t is a **time series**.

Time step Δt : It is the interval between consecutive data points and is often consistent throughout the series such that : $y(t), y(t+\Delta t), y(t+2\Delta t), \ldots$

Examples



A time series graph of the population of the United States from 1900 to 2000.C.K. Taylor



Electricity consumption in France.
from july 2019 to january 2023. ENEDIS

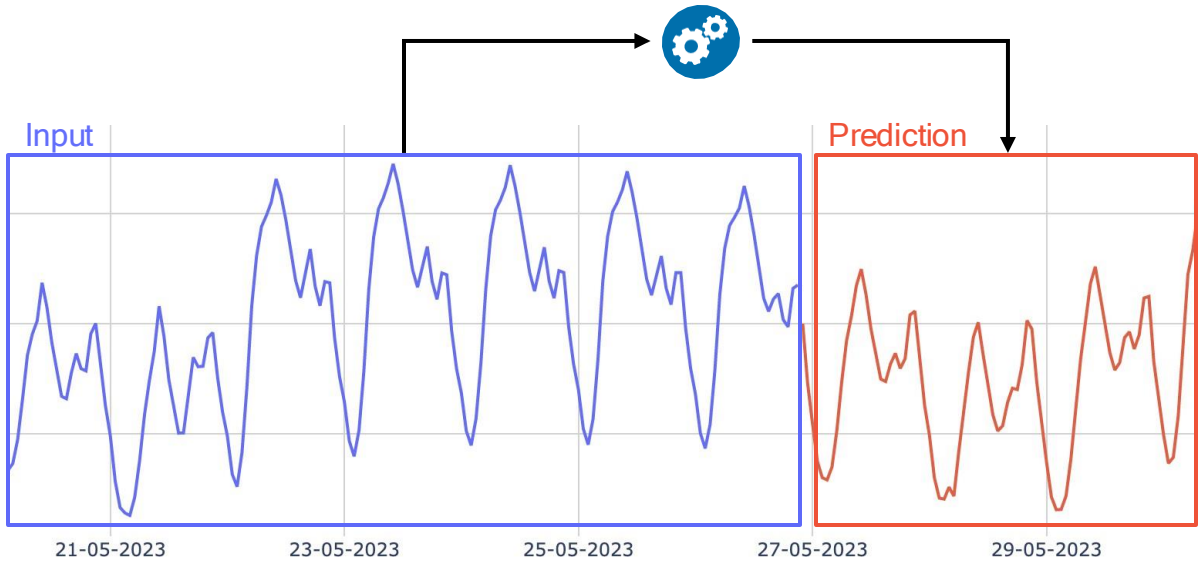
ECG

TIME SERIES USE CASES: FORECASTING

Predicting future outcomes based on historical data

- Use historical data to estimate future events. Exogenous data can be added. Prediction models can be univariate (predict a single variable) or multivariate (predict multiple variables).

- Methods: statistical techniques such as ARIMA, exponential smoothing methods, or machine learning methods such as recurrent neural networks.



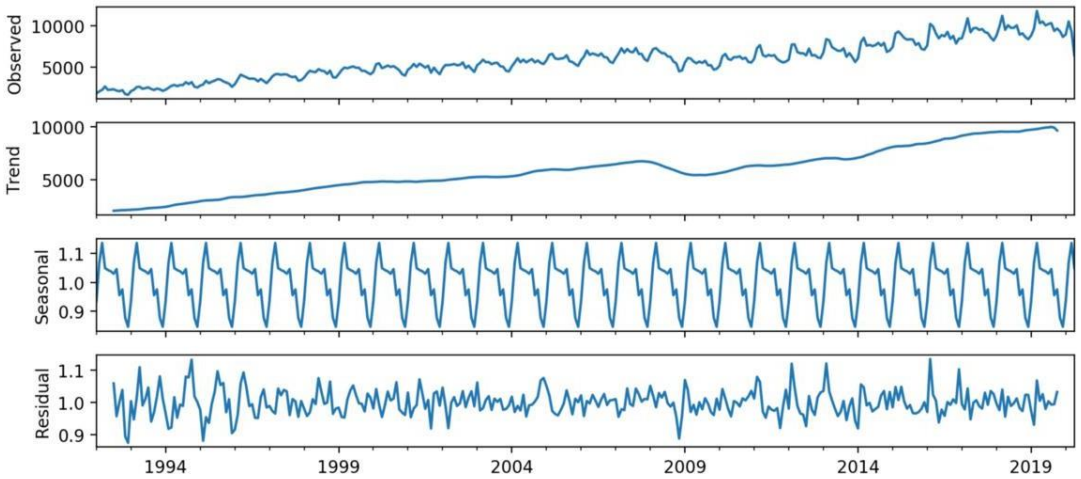
USE CASES:

- **Sales Forecast:** Forecasting future sales based on historical sales data.
- **Weather Forecast:** Prediction of future weather conditions based on historical weather data.
- **Energy Demand Forecasting:** Forecasting future energy demand based on historical usage patterns.

TIME SERIES USE CASES: DECOMPOSITION AND TREND ANALYSIS

Decomposing time series into its constituent parts to identify and understand underlying patterns and directions over time

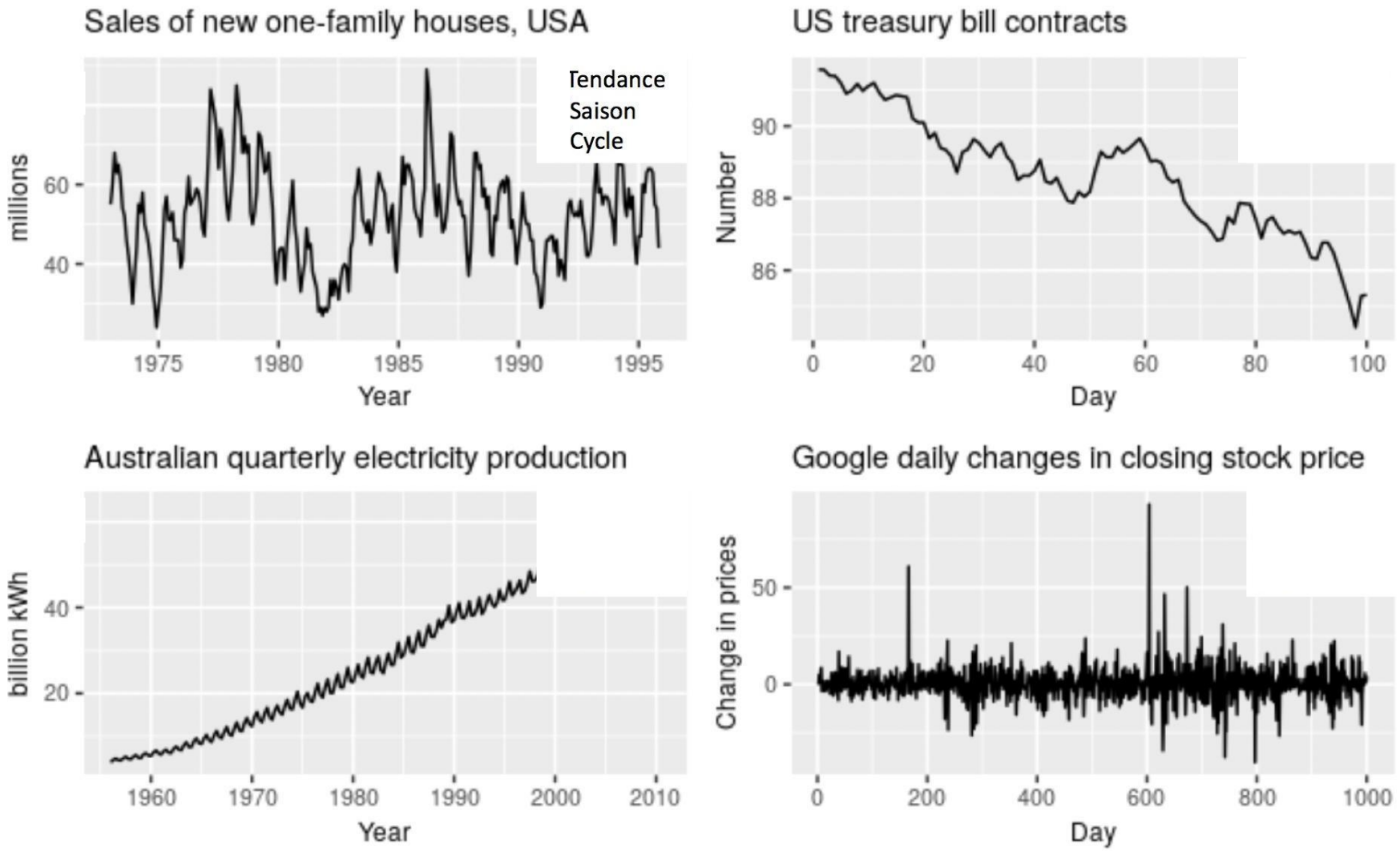
- Decomposition is a technique for separating a time series into several components, usually trend, seasonality, and residual.
- Time series trend analysis is the process of identifying and understanding patterns or general directions in the data over time.



CAS D'USAGES :

- **Market Research:** Identifying Changes in Consumer Behavior
- **Environmental monitoring:** Identify changes in environmental data, such as temperature or pollution levels
- **Sales Analysis:** Understanding the Factors Behind Sales Evolution

DECOMPOSITION OF TIME SERIES: TREND, SEASONALITY AND CYCLE

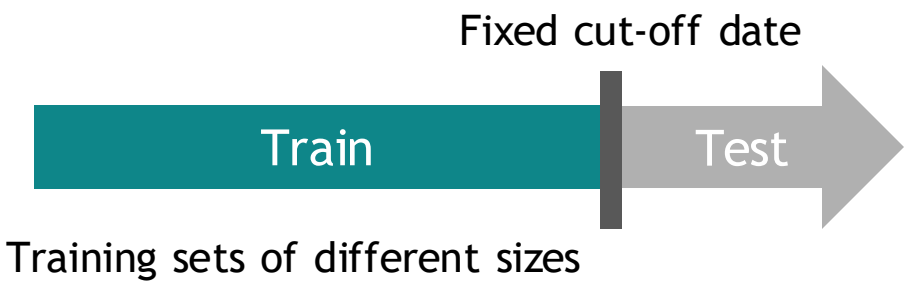


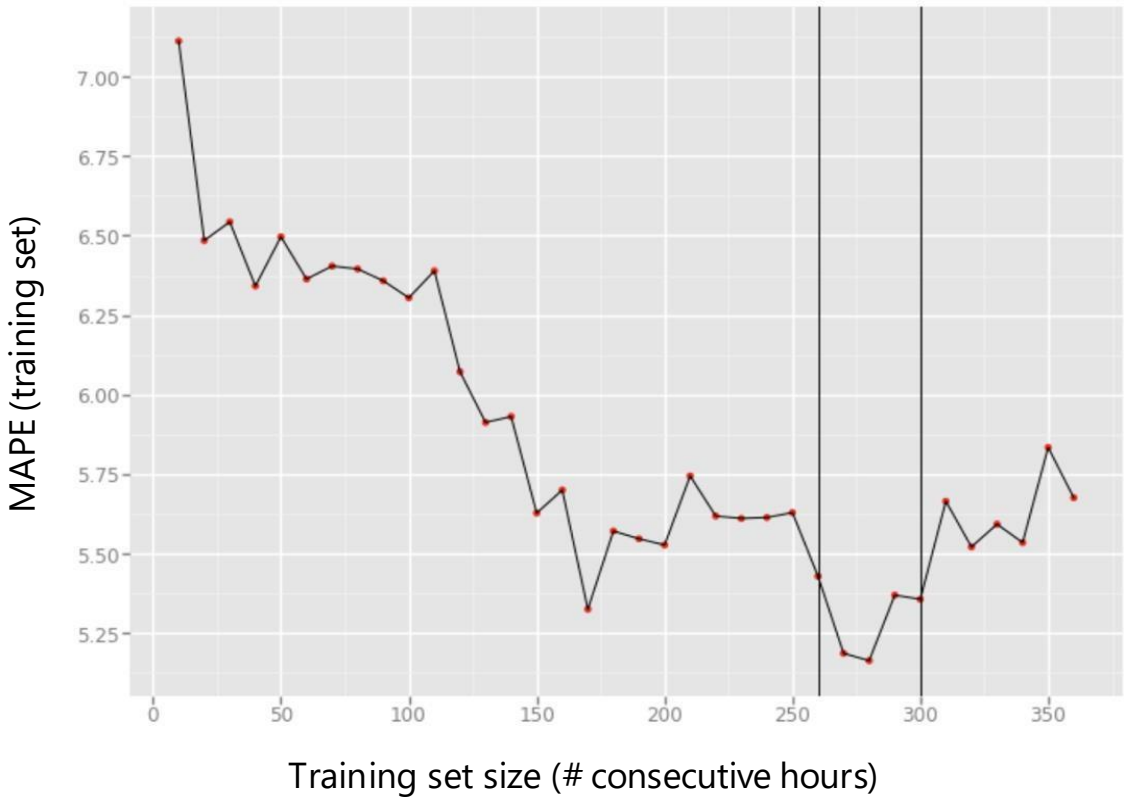
Some Key Concepts for TimeSeries

- CrossValidation in TimeSeries, and chosing training et test sets
- Lag
- AutoCorrelation, Partial Autocorrelation and PACF
- Stationarity
- ADF Test

SELECTION OF THE TRAINING AND TESTING SETS SIZES

Optimal size of the training set





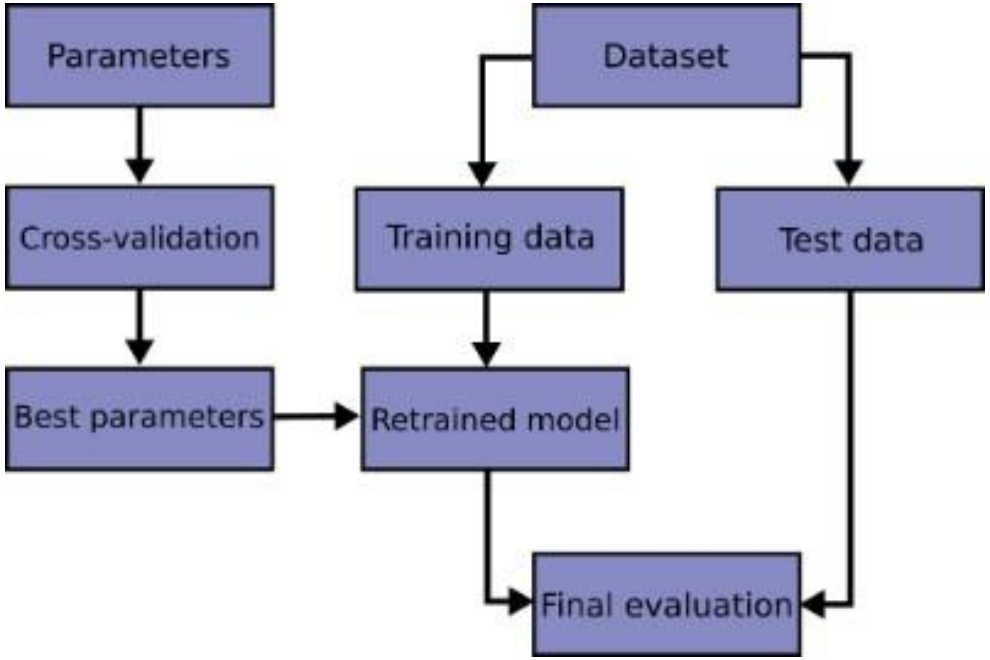
Choosing the right data for our **training** and **testing** sets is crucial for modeling the dynamics of interest effectively.
The **training set** should represent the underlying **processes** and **patterns** that we aim to understand and predict

CROSS VALIDATION IN TIME SERIES

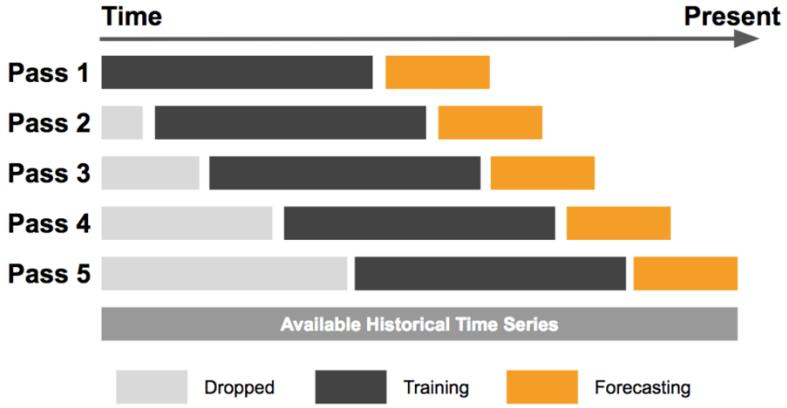
Why use cross validation?

- Optimise the model **hyperparameters**
- Avoid **overfitting**

Cross validation workflow



Sliding Window cross-validation



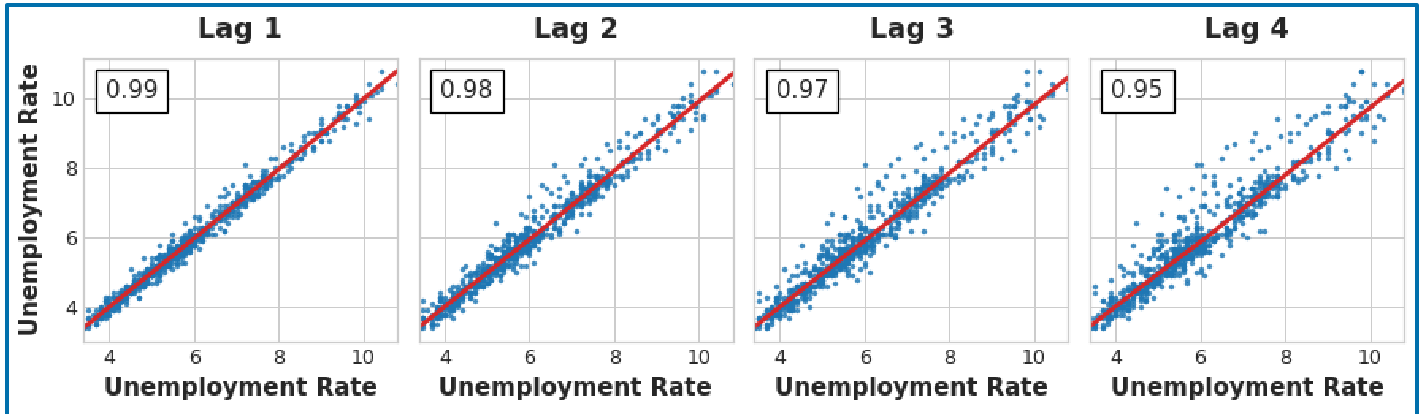
Increasing Window cross-validation

LAG IN TIME SERIES

Definition

- **Lag** is the time interval or the number of observations between a data point and its past values. When we talk about the kth lag of a series at the instant t, we therefore talk about y_{t-k}.

	y	y_lag_1	y_lag_2
Date			
1954-07	5.8	NaN	NaN
1954-08	6.0	5.8	NaN
1954-09	6.1	6.0	5.8
1954-10	5.7	6.1	6.0
1954-11	5.3	5.7	6.1



Lag plot of a time series shows its values plotted against its lags.

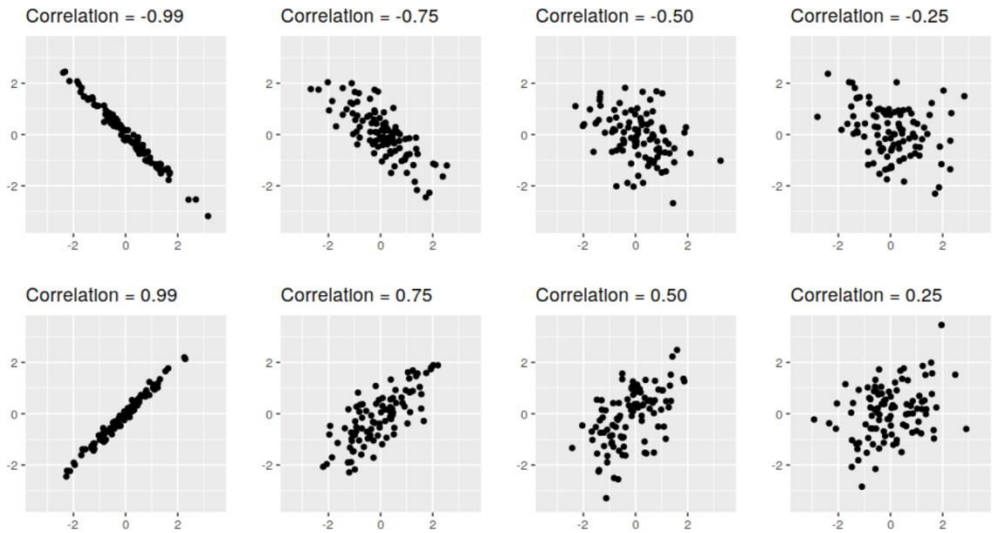
But how to determine which lags are importants ?

MEASURING THE INFLUENCE OF THE PAST VALUES: CORRELATION VS AUTOCORRELATION

Definition

Correlation measures the strength of a linear relationship between two variables:

$$r = \frac{\sum(x_t - \bar{x})(y_t - \bar{y})}{\sqrt{\sum(x_t - \bar{x})^2}\sqrt{\sum(y_t - \bar{y})^2}}.$$



Example of the correlation plot between two variables

Definition

Autocorrelation is based on the same idea but instead of comparing two different variables it compares a time series and its **kth-lag**:

$$s_k = \frac{1}{n}\sum_{i=1}^{n-k}(x_i - \bar{x})(x_{i+k} - \bar{x}) = \frac{1}{n}\sum_{i=k+1}^{n}(x_i - \bar{x})(x_{i-k} - \bar{x})$$

The ACF or correlogram is the representation of the autocorrelation for every lag: $\rho_k = \frac{s_k}{s_0}$

MEASURING THE INFLUENCE OF THE PAST VALUES: THE PACF – PARTIAL AUTO CORRELATION FUNCTION

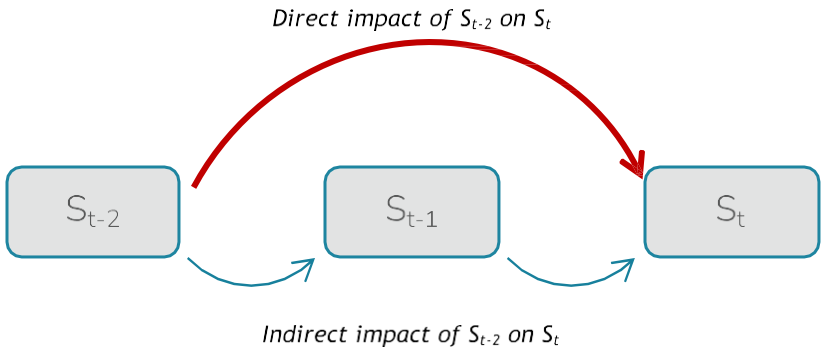
Definition

The partial autocorrelation function (PACF) measures the relationship in x_t and x_{t-k} after removing the effects of intermediate lags (lag 1, 2, …, k-1). Thus the correlations are isolated and the information already exploited is not reused.

- We denote r_k the k-ranked partial autocorrelation, which defines the coefficient of x_{i-k} in the linear regression on variables $1, x_{i-i}, \cdots$, such that: $x_i = \alpha_0 + \sigma_{j=1}^{k-1} \alpha_j x_{j-1} + r_k x_{i-k} + \varepsilon_i$
- This implies that the partial autocorrelation is the correlation between x_i and x_{i-k} once all the ranks between $x_{i-k+1}, \cdots, x_{i-1}$ are defined and removed from the equation.
- k-ranked partial autocorrelation: $x_i = \alpha_1 x_{i-1} + \alpha_2 x_{i-2} + \cdots + \alpha_{k-1} x_{i-k+1} + U,$

$$x_{i-k} = \beta x_{i-1} + \beta_2 x_{i-2} + \cdots + \beta_{k-1} x_{i-k+1} + V,$$

$$r_k = corr(U, V)$$



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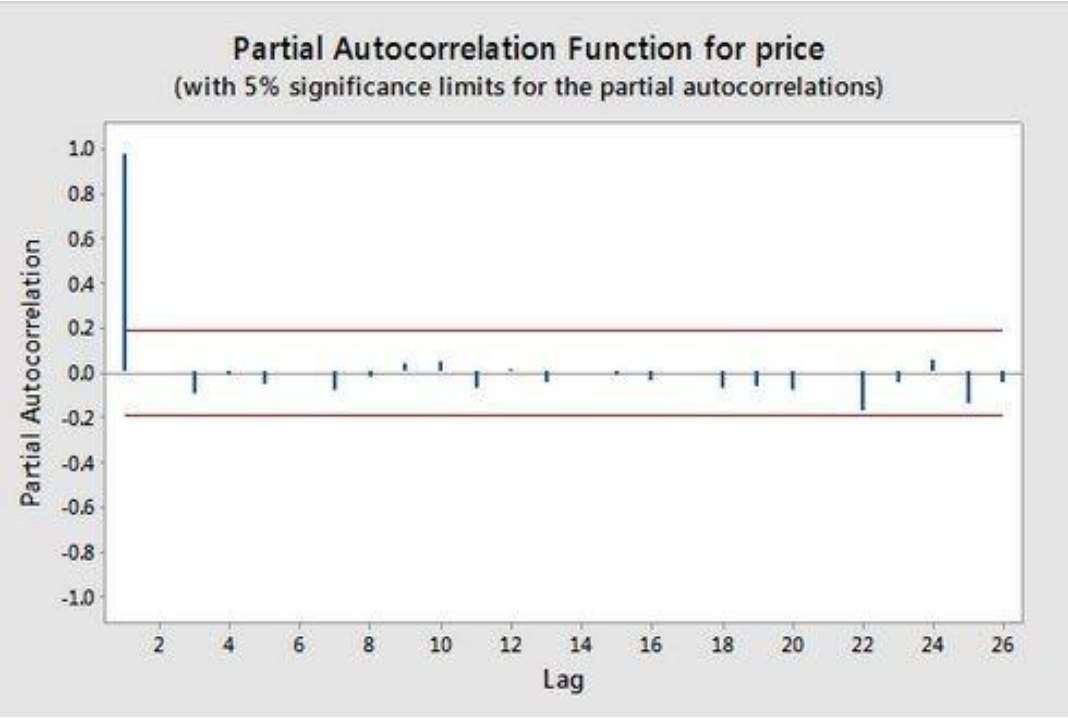
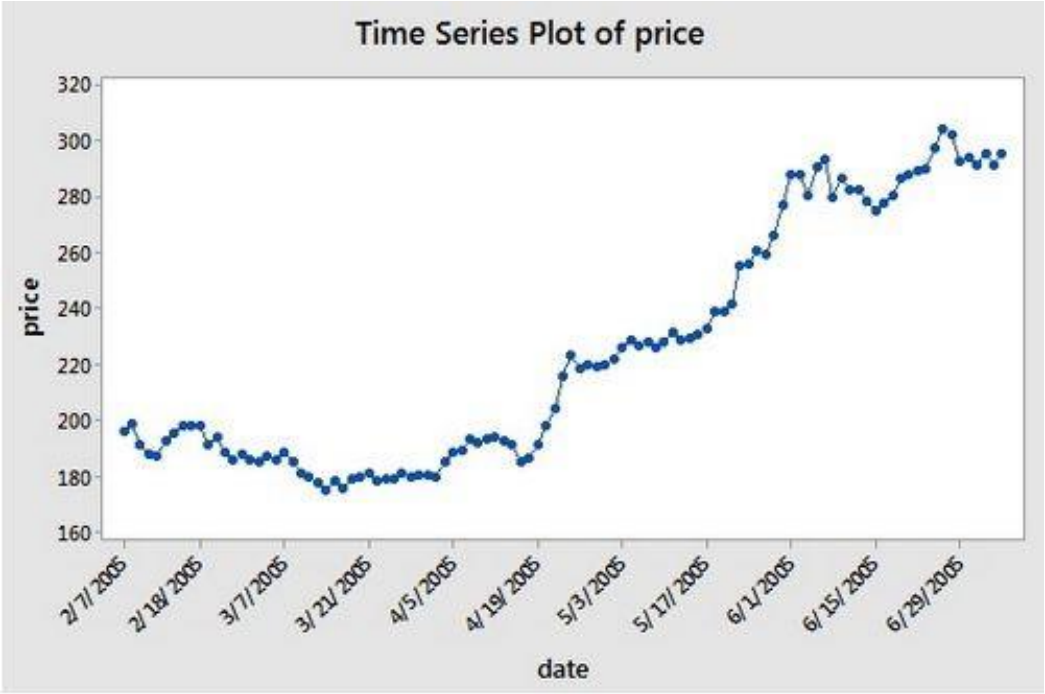
Challenges in Time Series Analysis - Autocorrelation
Lag of k: the k-th point previous value in the past





Google Stock price

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Challenges in Time Series Analysis - Autocorrelation
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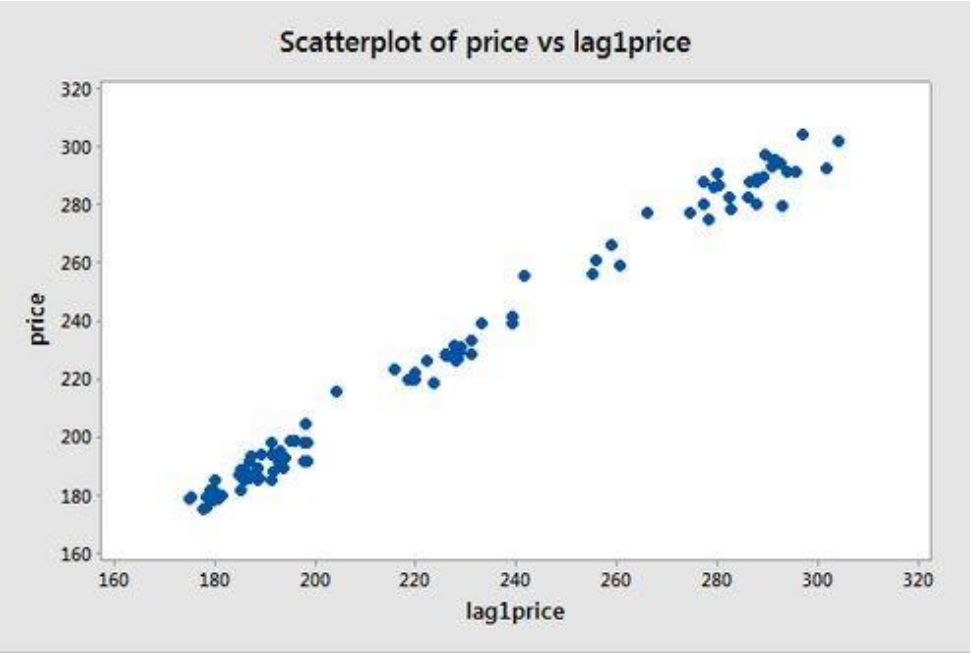
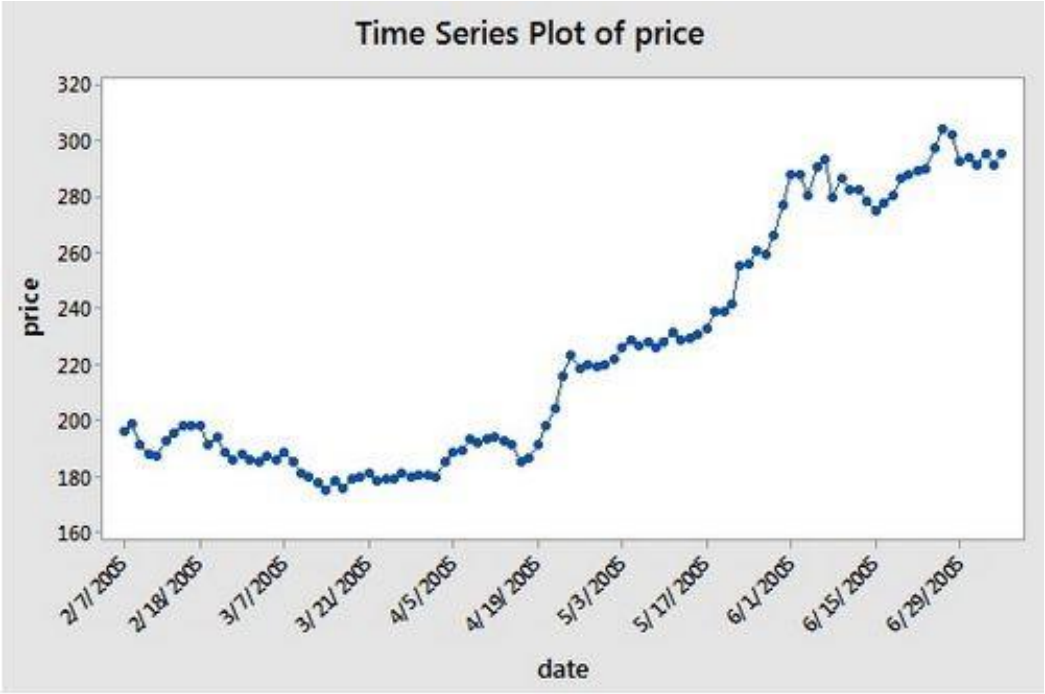




Google Stock price

$$y_t = \beta_0 + \beta_1 y_{t-1} + \epsilon_t$$

STATIONARITY

A stationary time series is a series whose properties do not depend on when the series is observed.

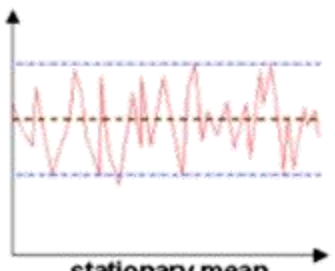
Definition

- A process, $\{y_t\}$, is **strictly stationary** if for each k, t and n, the distribution of $\{y_t, \dots, y_t + k\}$ is the same as the distribution of $\{yt + n, \dots, yt + k + n\}$
- $\{yt\}$, is **2nd order stationary** (or **weakly stationary**) if $E[y_t]$, $E[(y_t - E[y_t])^2]$, and $cov(y_t, y_{t+k})$ do not depend on t.

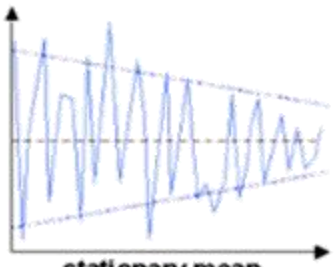
When talking about stationarity, we most often refer to the weak stationarity.

When is a Time serie stationary ?

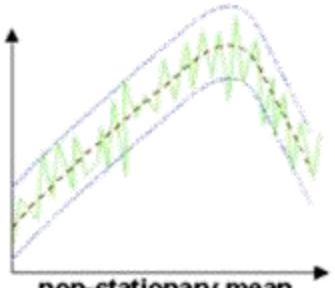
- **Time series with trend or seasonality are not stationary** – they will affect the value of the series at different times.
- **A white noise is stationary** - no matter when you observe it, it looks the same over time. It is even a strong stationarity.
- **Some cases can be confusing** - a time series with cyclical behavior is stationary, because cycles are not regular, unlike seasonality.



stationary mean
stationary variance



stationary mean
non-stationary variance



non-stationary mean
stationary variance

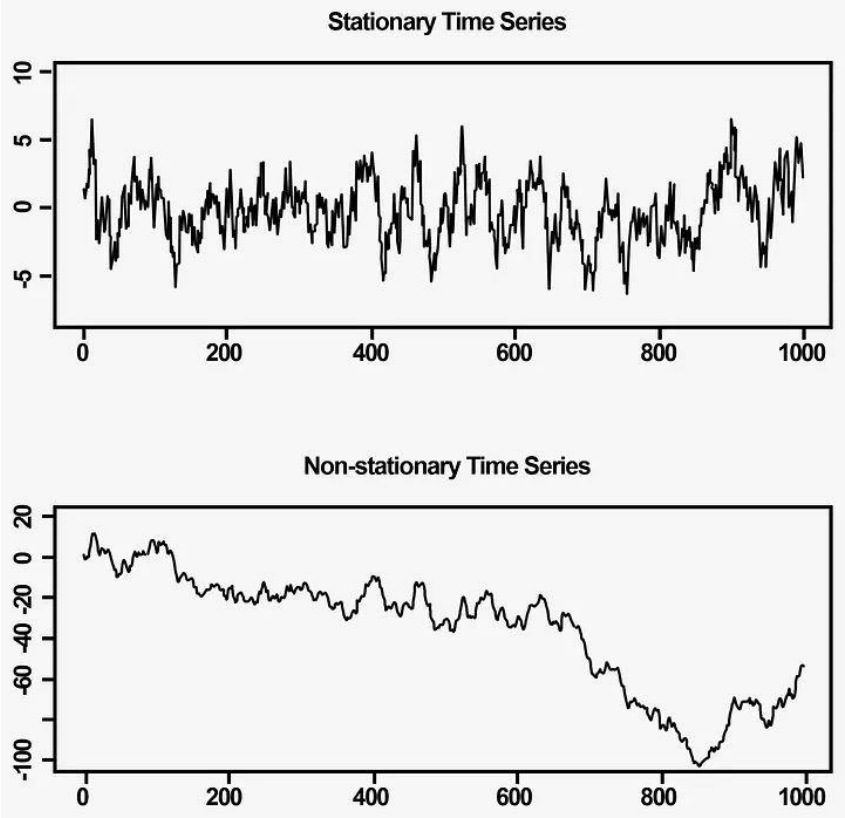
Machine Learning - Time Series

Challenges in Time Series Analysis - Non-stationarity problem

Stationarity: the statistical properties of the data points are consistent and do not depend on the time at which they are observed

Problem: Makes modeling and forecasting difficult because the assumptions of many statistical methods (like linear regression) are violated.

Solution: Differencing to remove trends and seasonality



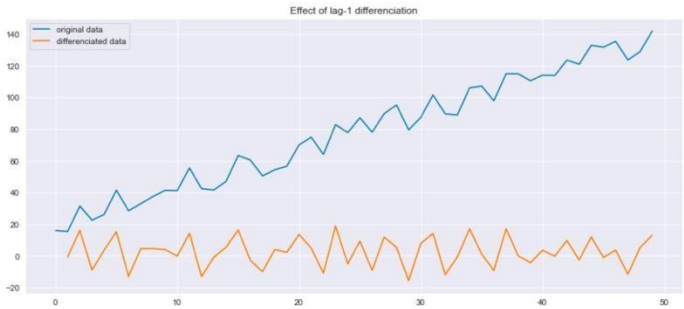
HOW TO MAKE SERIES STATIONARY?

- Remove the trend with differenciation:

$$Y_t = X_t - X_{t-1}$$

The simplest form of differencing is to subtract the previous observation from the current observation.
Example with a linear trend : $X_t = a + bt + \varepsilon_t \rightarrow Y_t = X_t - X_{t-1} = \mathrm{b} + (\varepsilon_t - \varepsilon_{t-1})$

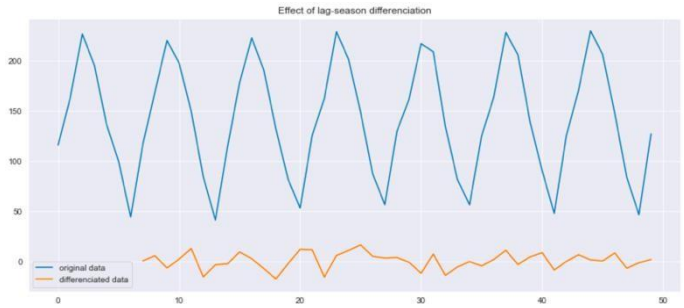
Differencing twice enables to remove a quadratic trend, etc.



- Remove the seasonality with seasonal differenciation:

$$Y_t = X_t - X_{t-s}$$

If the time series has a seasonal component, you might need to take the difference with a lag that matches the seasonality period.
Example : $X_t = a + S_t + \varepsilon_t \rightarrow Y_t = X_t - X_{t-s} = (\varepsilon_t - \varepsilon_{t-1})$ as $S_{t+s} = S_t$



- Variance stabilization with the Box-Cox transformation:

The value $\boldsymbol{\lambda}$ is chosen by seeing which value best approximates the transformed data to the normal distribution.

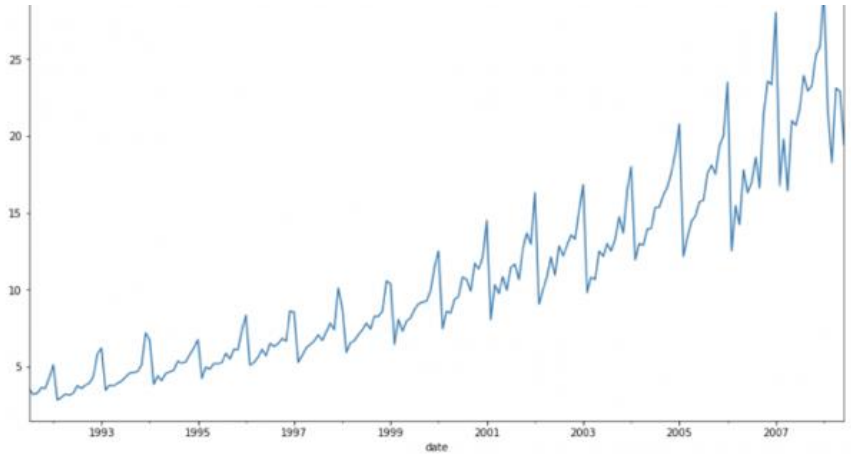
$$w_t = \begin{cases} \log(y_t) & \text{if } \lambda = 0; \\ (y_t^{\lambda} - 1)/\lambda & \text{otherwise.} \end{cases}$$

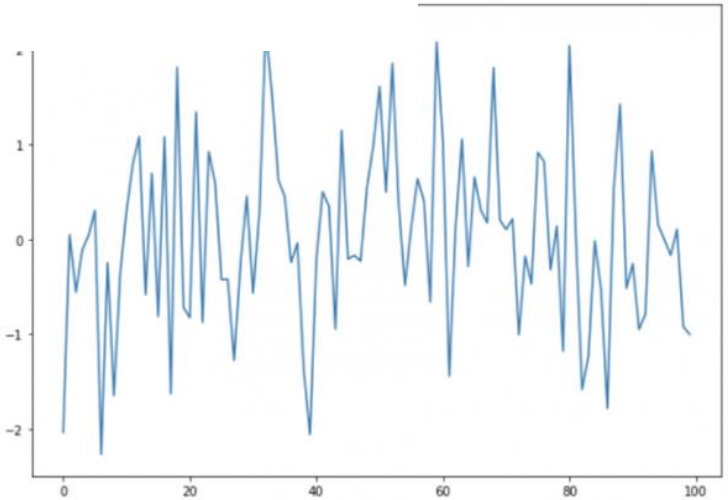
STATISTICAL TESTS

Statistical tests are essential in time series analysis for diagnosing the underlying properties of the data, such as stationarity, trends, and autocorrelation, which are critical for selecting appropriate models and ensuring accurate forecasts and analyses

Test	Description
Augmented Dickey-Fuller (ADF) Test	This test is used to check for the presence of a unit root in a time series, helping to determine if the series is non-stationary.
Granger Causality Test	This test is used to determine if one time series can predict another. It's a hypothesis test to determine if past values of one variable help predict the future value of another.
Ljung-Box Q Test	This test is used to determine if a time series is randomly distributed (i.e., no autocorrelation). It's commonly used to test the residuals of a time series model.
Cointegration Test (e.g., Johansen Test)	Cointegration tests are used to determine if a linear combination of non-stationary time series is stationary, indicating that the series move together in the long run.

$$y_t = c + \beta t + \alpha y_{t-1} + \phi_1 \Delta Y_{t-1} + \phi_2 \Delta Y_{t-2} .. + \phi_p \Delta Y_{t-p} + e_t$$





Source : https://www.machinelearningplus.com/time-series/augmented-dickey-fuller-test/

FEATURE ENGINEERING: USING LAGS AS FEATURES

Lag Features are **previous time points** in the serie used to predict future time points.

All lags are not equally informative: the lags to include as features can be determined based on **domain knowledge** and **autocorrelation** analysis.

	y	y_lag_1	y_lag_2
Date			
1954-07	5.8	NaN	NaN
1954-08	6.0	5.8	NaN
1954-09	6.1	6.0	5.8
1954-10	5.7	6.1	6.0
1954-11	5.3	5.7	6.1

RISKS:

Including too many lags can introduce **overfitting** :

- Alternatively, **rolling window statistics** such as mean, max, min can be informative.
- **Regularisation** methods such as l1 or l2.

Be careful to data leakage

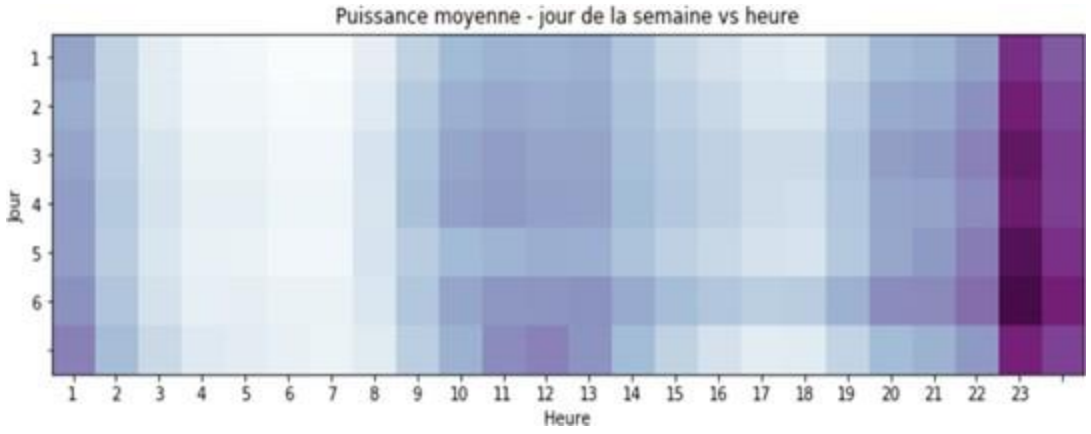
BEFORE CHOOSING THE MODEL: FEATURE ENGINEERING

The date often has a great impact on the prediction (**year**, **quarter**, **month**, or even sometimes the **day** of the week). They witness the possible seasonality in the data for example.

Thus it is often a great source of information that one should integrate in the model's features.
We often decompose the date first (year/month/day/hour) as each component brings its own piece of information.
Feature transformations can also be done (aggregation, encoding, etc.).

How to integrate seasonality?

- Insert lags with an offset equal to that seasonality
- Add columns with the season
- Add columns that group similar values together
- Etc.

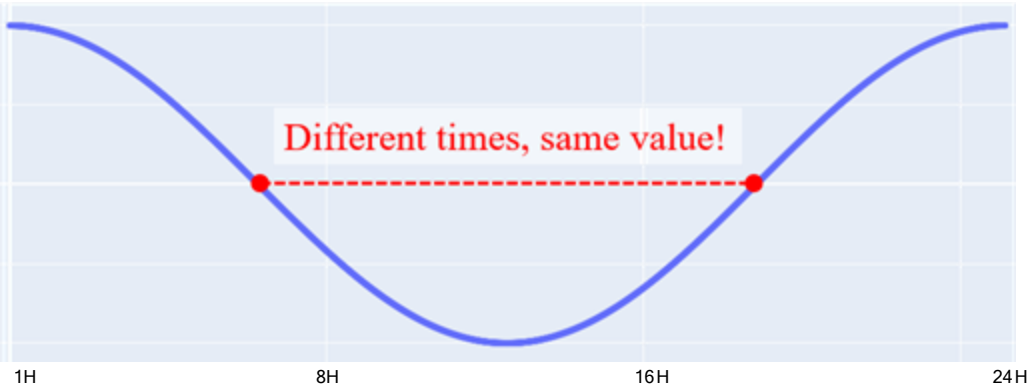


BEFORE CHOOSING THE MODEL: FEATURE ENGINEERING

Hours of the day, days of the week, months of the year occurs in cycles : proper encoding preserves the **continuity** and **natural order** of cycles (12am and 1am are close to each other).

1. Convert to 1 dimension with sin/cos functions:

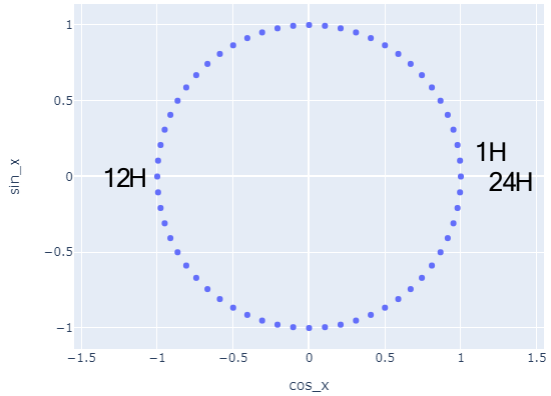
- **Methods**: run the cosine function after normalizing the variable between 0 to 2π.
- **Limits**: Two different times would get the same value



2. Convert to 2 dimensions – map cyclical variables to a unit cirlcle:

- **Methods**:

$$x = \sin\left(\frac{a \times 2\pi}{\max(a)}\right) \qquad y = \cos\left(\frac{a \times 2\pi}{\max(a)}\right)$$



HOW TO DECOMPOSE A TIME ITH SINGULAR SP NALYSES

Time series decomposition allows for the isolation and ana nponents such as trend, seasonal facilitating a clearer understanding and more accurate forecasting of the unde e data.

Methodology

We define our TimeSeries X :

$$\{X(t): t = 1, \dots, N\}$$

We compute the **MxM** *lag-covariance matrix* ***Cx***

$$C_{ij} = \frac{1}{N - |i - j|} \sum_{t=1}^{N-|i-j|} X(t)X(t +$$

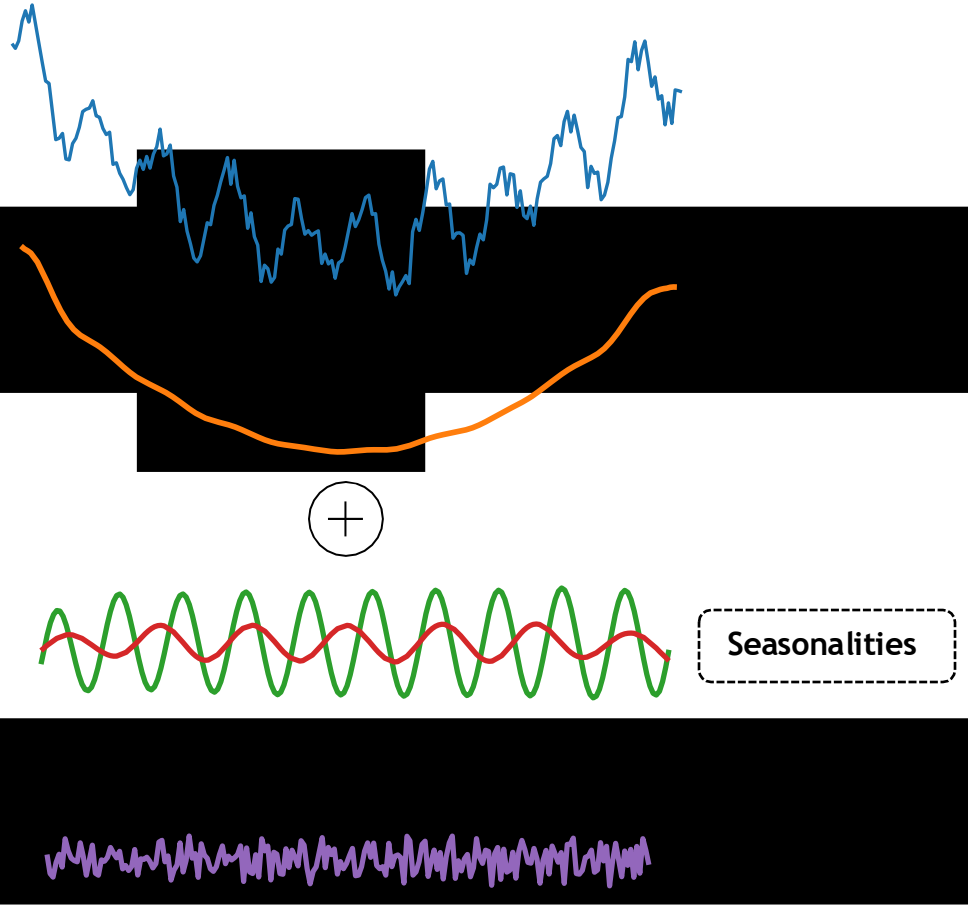
Lagged tr

Diagonalisation of the lagged covariance matrix

- Singular values : Eigen values
- Empirical orthogonal function (EOF) : Eigen ve

We project the TimeSeries X onto eigen vectors ***Ek*** *s principal component has the following form*

$$A_k = \sum_{j=1}^{M} X(t + j - 1)E_k(j)$$



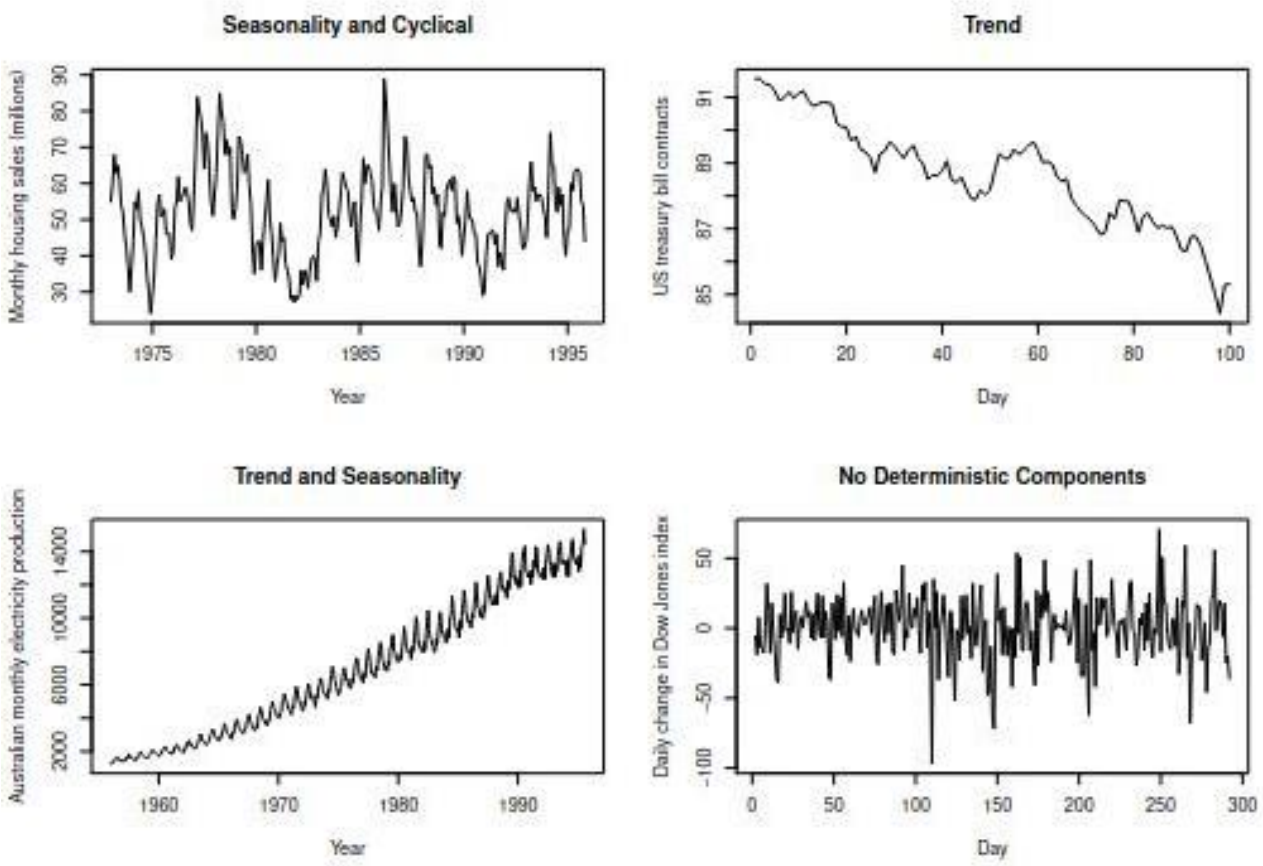
Machine Learning - Time Series

Time Series decomposition

Decomposition: **Trend**, **Seasonality**, **Residual**

Tools for decomposition (STL, X-13ARIMA-SEATS)
In Statsmodel: statsmodels.tsa.seasonal.STL

statsmodels.tsa.x13.x13_arima_analysis

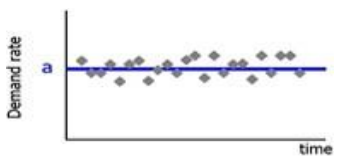


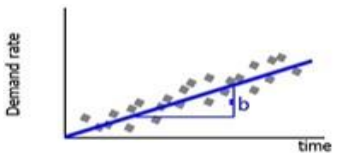
FORECAST APPROACHES – TIME SERIES

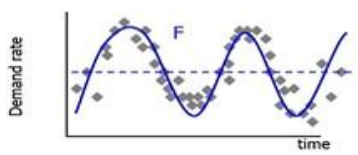
"Past Predicts the future" - Building a model

A forecast model must translate as much as possible the factors the influence the demand.
Thus, the different components can be combined in different ways :

Forecast Model = **Level (a)** +/x **Trend (b)** +/x **Seasonality (F)** + **Random fluctuations (ε)**







Examples :

Level Model	$x_t = a + \varepsilon_t$
Level - Trend Model	$x_t = a + bt + \varepsilon_t$
Mix Level-Seasonality Model	$x_t = aF_t + \varepsilon_t$
Mix Level-Trend-Seasonality Model	$x_t = (a+bt)\,F_t + \varepsilon_t$

Machine Learning - Time Series

Regression Models for Time Series
Popular (classic) model: Autoregressive Integrated Moving Average (**ARIMA**) model

- The AR part involves using the dependency between an observation and a number of lagged observations.
- The I part represents the differencing of raw observations to make the time series stationary.
- The MA part models the error of the time series as a linear combination of error terms from the past.

To use it: statsmodels.tsa.arima.model.

Generalized autoregressive conditional heteroskedasticity (**GARCH**) models aim to model the conditional volatility of a time series (very useful in finance)
To use it: https://github.com/bashtage/arch
To go deeper:
https://machinelearningmastery.com/develop-arch-and-garch-models-for-time-series-forecasting-in-python/

In practice, other regression models can apply, on features + past target values!

MODELING A SERIES WITH AN AR PROC[redacted]

AR(p) : AutoRegressive of order p>0

- In an AR model, the value of the time series at a particular time point [redacted] to be a **linear combination** of its **previous values** plus a stochastic term (which is an error term, often a[redacted] be white noise).
- **Hypothesis :**
  - **AR(p)** values depend on its **p-previous** values (lags)
  - An AR processus must be **stationary**

$$\forall t \in \mathbb{Z}, \qquad X_t = \sum_{i=1}^{p} \varphi_i X_{t-i} + \varepsilon_t \qquad \varphi_p \neq 0$$

X_t is modeled by a linear relation with its lags

The partial autocorrelations of $(X_t)_{t\in\mathbb{Z}}$ cancel out from the rank p+1
The autocorrelations quickly drop to 0.

MODELING A SERIES WITH AN MA PROCESS

MA(q) : Moving Average of order q>0

- The MA model is defined by the **linear relationship** of a time series with cur ast **white noise** error terms.
- **Hypothesis:**
  - The deviation **MA(q)** depends on the **q-previous** deviations
  - An MA processus must be **stationary**

$$\forall t \in \mathbb{Z}, \qquad X_t = \varepsilon_t + \sum_{i=1}^{q} \theta_i \varepsilon_{t-i} \qquad (\theta_1, \ldots, \theta_q \neq 0$$

X_t is modeled by a linear relation of decorrelated perturbations (a white noise and its past)

The autocorrelations of $(X_t)_{t\in\mathbb{Z}}$ cancel out from the rank q+1

ARMA PROCESSES ARE A COMBINAT[redacted]OF A[redacted]AND MODEL

ARMA (p,q) : AutoRegressive Moving Average, also called Box-Jenk[redacted]of orde[redacted]

- ARMA(0,q) = MA(q)
- ARMA(p,0) = AR(p)
- An ARMA process must be **stationary**

$$\forall t \in \mathbb{Z}, \qquad X_t = \boxed{\sum_{i=1}^{p} \varphi_i X_{t-i}} - \varepsilon_t + \boxed{\sum_{i=1}^{q} \theta_i \varepsilon_{t-i}} \qquad \begin{cases} (\varphi_1, \dots, \varphi_p) \in \mathbb{R}^p \\ \varphi_p \neq 0 \end{cases}$$

X_t is modeled by both an AR and an MA process

FROM ARMA TO ARIMA: INTEGRATE A TREND COMPONENT TO THE MODELING

One of the limitations of ARMA models is the **stationarity** condition.

In many situations, time series can be viewed as composed of two components, a **non-stationary trend series** and a **stationary zero mean** series:

$$X_t = \mu_t + Y_t, \text{with } Y_t \text{ an ARMA process and } \mu_t \text{ a trend component}$$

Strategies :

1. **Detrending**: subtraction with a trend estimate and residuals management.

$$Y_t' = X_t - \mu_t'$$

2. **Differencing**:
   We can model the trend as a random component as

$$\mu_t = \delta + \mu_{t-1} + \varepsilon_t$$
$$\nabla X_t = X_t - X_{t-1} = \delta + \varepsilon_t + (Y_t - Y_{t-1}) = \delta + \varepsilon_t + \nabla Y_t$$

∇ is defined as the first difference and can be extended to higher orders ∇^i.

FROM ARMA TO ARIMA: DIFFERENCING THE SERIES

ARIMA (p,d,q) : Autoregressive Integrated Moving average of order p, d and q.

- **p** : number of autoregressive terms
- **d** : number of non-seasonal differences to reach stationarity
- **q** : number of lag errors in the prediction equation

The first difference eliminates **a linear trend**.
A second difference, i.e. the difference of the first difference, can eliminate a **quadratic trend**.

With Backshift operator : $X_{t-1} = BX_t$

A process X_t is called ARIMA (p,d,q) if :

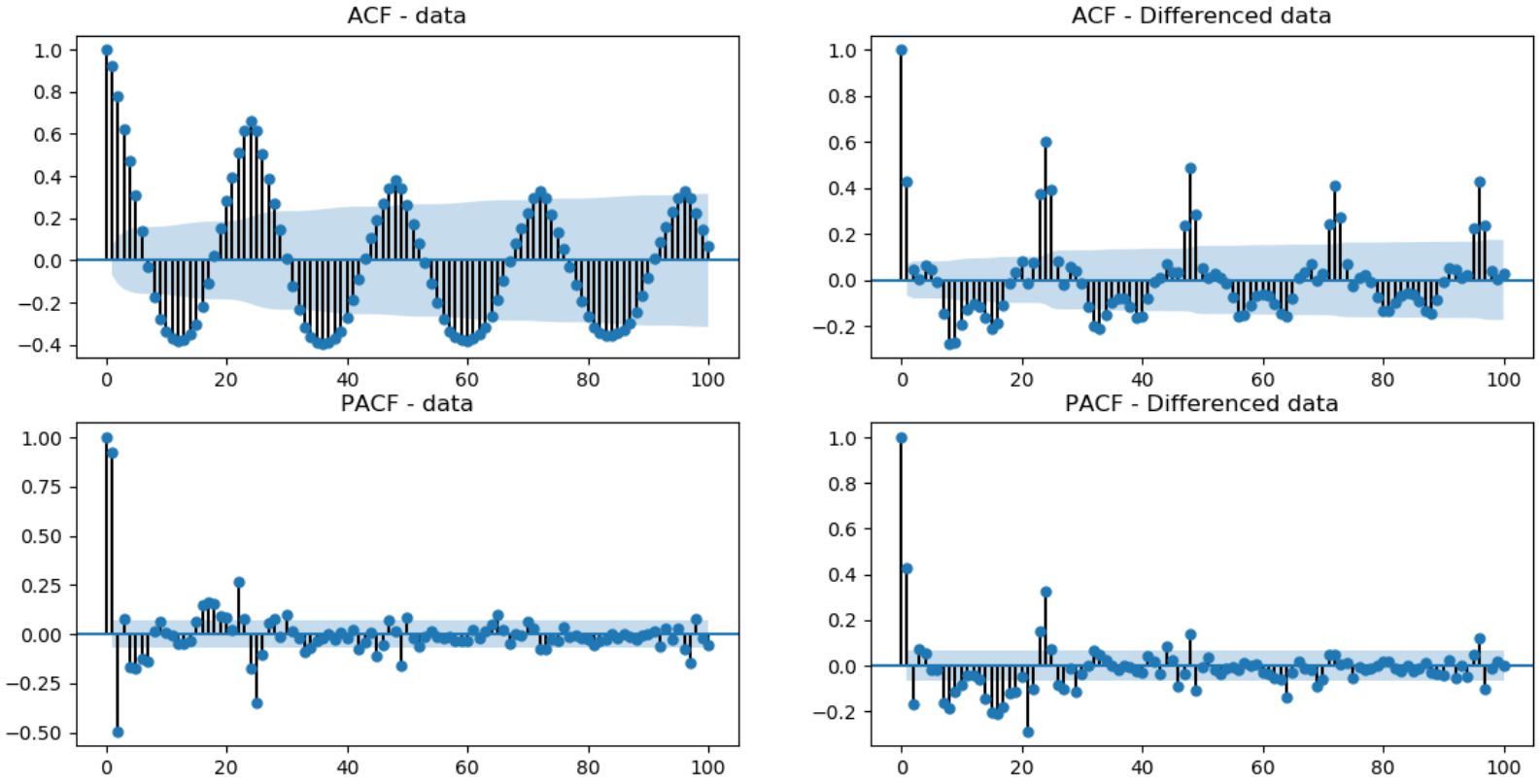
$$\boxed{\nabla^d X_t = (1-B)^d X_t \text{ is } ARMA(p,q)}$$

In general, ARIMA(p,d,q) can be written:

$$\boxed{\begin{aligned} &\nabla^d = (1-B)^d \\ \Phi(B)\nabla^d X_t = \Theta(B)\varepsilon_t \qquad &\Phi(B) = 1 - \varphi_1 B - \cdots - \varphi_p B, \qquad \varphi_p \neq 0 \\ &\Theta(B) = 1 - \theta_1 B - \cdots - \theta_q B, \qquad \theta_q \neq 0 \end{aligned}}$$

Machine Learning – Time Series

ACF (gives q) vs PACF (gives p)



Machine Learning - Time Series

Tools

Sklearn/Statsmodels

Recurrent Deep learning models (RNN, LSTM, Transformers) are well suited for time series prediction.

AutoML tools like Facebook Prophet GitHub - facebook/prophet: Tool for producing high quality forecasts for time series data that has multiple seasonality with linear or non-linear growth.

Time Series specific package:

- **Darts** https://pypi.org/project/darts/